

THE INTERN DIARIES

Intern Report

Monthly Sales Trend

WEEK 3 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

Retail_Sales (RETAIL COMMERCE)

1. How do you guarantee the months plot in calendar order (Jan→Dec) and not alphabetical order (Apr, Aug, Dec, Feb...)?

Answer: Never group or sort on the month *name*. Create a sortable **month key** with `=TEXT([@Clean_Date],"YYYY-MM")`, which yields zero-padded strings like 2023-01, 2023-02 ... 2023-12 that sort in true chronological order. Keep a *separate* display label `(=TEXT(DATEVALUE($A2&"-01"),"mmm-yyyy") → Jan-2023)` purely for the chart's axis, so the line reads nicely while the underlying order stays correct. The YYYY-MM column is the sort spine; the label is decoration.

2. The CSV dates arrived as text — how do you catch that before it silently breaks your totals, and how do you fix it?

Answer: We test first, but don't assume: `=ISNUMBER(B2)` returns **FALSE** if a date is text, and `=SUMPRODUCT(--(ISTEXT(Sales[Date])))` counts how many rows failed. Fix with a guarded helper column `Clean_Date =`
`=IFERROR(DATEVALUE([@Date]),[@Date])` (converts text dates, passes real dates through), then confirm `=COUNT(Sales[Clean_Date])` equals the full row count (1,000). A text date that slips through lands in *no* month bucket, so the reconciliation in Q4 would also flag it.

3. Why a line chart for this task instead of a bar chart of monthly totals?

Answer: The question being asked is "*how does sales change over time?*" — a **trend** question — and line charts are the correct type for change across an ordered time axis, because the connecting segments make rises, dips and seasonality visible at a glance. A bar chart compares discrete categories; it can show 12 monthly totals but it doesn't emphasize the *shape* of movement between them. So: line with

markers, titled Monthly Sales Trend — Retail Sales Dataset (2023), X-axis Month, Y-axis Total Sales (\$).

4. How do you prove the monthly rollup neither dropped nor double-counted a single transaction?

Answer: Reconcile the breakdown against the control total on a Validation tab, using tolerance (not =) because money is rounded:

- Sales: =IF(ABS(SUM(B2:B13) - SUM(Sales[Total Amount])) < 0.01, "✓
MATCH", "✗ CHECK")
- Counts: =IF(SUM(C2:C13) = COUNTA(Sales[Transaction ID]), "✓
MATCH", "✗ CHECK") (must equal 1,000)
- Plus the type check from Q2. If the 12 monthly figures don't sum to the grand total, a date failed conversion and vanished from every bucket — the check catches it. Three independent proofs (type, sum, count) make the table trustworthy.

5. What three measures make up the monthly summary table, and how is each computed?

Answer: One row per month (2023-01...2024-01), with:

- **Total Sales** = =SUMIFS(Sales[Total Amount], Sales[Month_Key], \$A2) → currency
- **Transactions** = =COUNTIFS(Sales[Month_Key], \$A2) → whole number
- **Avg Sale / Transaction** = =IFERROR(B2/C2, 0) → currency (the IFERROR guards a zero-transaction month)

All three key off Month_Key, so the table is fully formula-driven and recalculates if the source changes — the same structured-reference habit from Days 8/10/14, now applied to a time axis.